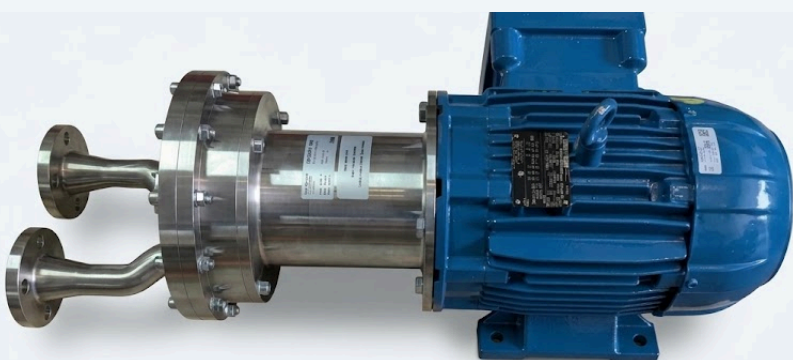




PRESSURIZED GAS RECIRCULATION

A sealed, high-pressure regenerative blower for a natural gas dryer

How we co-engineered a custom, zero-leak blower that runs inside a pressurized natural gas circuit, and why the customer is now ordering more.



DELIVERED UNIT

AT A GLANCE

Customer	European gas-treatment OEM (North American project)
Application	Gas recirculation on a natural gas drying unit
Flow	20 SCFM
Inlet pressure	12 bar(g) (174 psig)
Outlet pressure	~12.5 bar(g) (181 psig)
Motor	3 hp
Pressure test	16 bar (232 psi)
Sealing	Hermetic magnetic coupling, no shaft seal
In service	~18 months, still in operation
Status	Additional units requested

The challenge

The customer needed to move gas inside a closed, pressurized drying circuit. Off-the-shelf regenerative blowers are built for near-atmospheric service, and a conventional shaft seal would be a permanent leak path for natural gas.

- **Whole blower at system pressure:** the casing itself had to be a pressure-containing part.
- **Zero emissions** to atmosphere, with no dynamic seal to maintain.
- **Pressure-code discipline:** design and testing aligned with the customer's ASME practice.
- **Hazardous-area** motor and easy integration into existing skid piping.

Our solution

We engineered a dedicated blower around the customer's process, working with their engineering team through drawing approval.

- **Pressure-rated housing,** wall thickness calculated for the operating pressure and hydrostatically tested at 16 bar.
- **Magnetic drive** separating the process gas from the motor: no shaft seal, no leakage.
- **Tangential inlet/outlet** for simple pipe routing.
- **ANSI flanges** and a certified motor.
- Performance extended to a **higher differential pressure** than first specified, at the customer's request.



Results

~18 mo

in field service, and still running

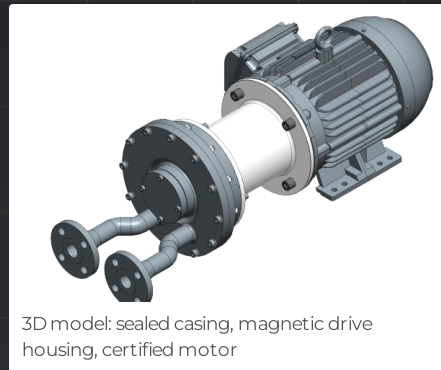
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shaft seals: gas stays inside the casing

Repeat

the customer is requesting additional units

Since commissioning, the blower has been operating on the customer's natural gas drying unit. The strongest proof of performance is the customer's decision to come back for more units of the same design for new projects.



From first unit to next generation

This project defined our pressurized-blower platform. Its main limit was the magnetic coupling, rated to 16 bar. The platform we are proposing for new gas applications keeps the same principles (sealed casing, magnetic drive, process-specific engineering) and pushes the pressure envelope much further.

	REFERENCE PROJECT	CURRENT PLATFORM
Magnetic coupling	Rated 16 bar	High-pressure coupling rated 63 bar
Housing	Pressure-rated casing	Billet aluminum, nickel coated
Shaft / bearings	—	316 stainless shaft, hybrid ceramic bearings
Motor	3 hp certified motor	Inverter duty, Class I Div 2 Group D
Flow range	Built for one application	One frame covers a full product range through speed control

Why it matters for gas processing OEMs

- **No leak path:** magnetic drive removes the shaft seal, the usual failure and emissions point.
- **Built for your pressure:** the casing is engineered and tested as a pressure part, not adapted after the fact.
- **Engineered with you:** connections, materials and motor are specified around your skid.

Have a pressurized gas application?

Let's review your operating conditions together.

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